```
openldap-2.4.19-1.fc12.i686
openldap-devel-2.4.19-1.fc12.i686
```

Installing the OpenLDAP client

To be able to query other LDAP servers, you will need to install the *openldap-clients* package:

```
[root@vbg-work ~]# yum install openldap-clients
```

With the client software installed, you can access other LDAP servers. There are quite a few LDAP servers on the Internet that allow anonymous querying, which implies querying a server without providing a user name and password, much like in anonymous FTP transactions where you can download files without providing a username/password for authentication.

To query a directory server conforming to the LDAP protocol, use the *ldapsearch* command.

A list of public LDAP servers can be found at: *www.keutel.de/directory/public_ldap_servers.html*. To test whether your OpenLDAP client software is installed correctly and is able to query remote LDAP servers, you may query any one of the listed servers. As an example, I chose to query the server x500.bund.de:

```
[vbg@vbg-work scratch]$ ldapsearch -x -h x500.bund.de \
-b "o=bund,c=de"
```

The *-h* option in the *ldapsearch* command arguments specifies the hostname to be queried. The *-b* option specifies the 'base' from which the search should begin and *-x* specifies 'simple authentication'.

Running the command displays a lot of output on the screen. Redirecting this output to a file will help us in understanding LDAP:

```
[vbg@vbg-work scratch]$ ldapsearch -x -h x500.bund.de \
-b "o=bund,c=de" > /tmp/ldapTGH-1.txt
ldap_bind: Success (0)
        additional info: Bind succeeded
[vbg@vbg-work scratch]$
```

The stderror output (which was not redirected by the above command) shows that binding (or authenticating and connecting) to the LDAP server was successful. Let us now look at the contents of this file, which contain the output of our query:

```
[vbg@vbg-work scratch]$ less /tmp/ldapTGH-1.txt
# extended LDIF
#
# LDAPv3
# base <o=bund,c=de> with scope subtree
# filter: (objectclass=*)
# requesting: ALL
```

The opening line shows us that the output is in the LDIF format. LDIF stands for LDAP Data Interchange Format. LDIF is used to represent LDAP data in text form. An understanding of LDIF is critical to be able to perform successful add/modify/delete/search operations on an LDAP server. LDIF files are useful for:

1. adding data to the LDAP directory,
2. modifying directory data,
3. deleting directory data, and
4. pretty much anything that has to do with visualisation of directory data.

The other lines also give us some more meaningful information. Note that all these are preceded with a #. This means that these are comments, not the results of our query. They just provide additional information. The other important information is the version of the LDAP protocol used for querying (v3), the base of the LDAP tree on which the search operation was performed, and filters applied to refine search results.

A detailed explanation of this is beyond the scope of this article. Maybe in a future article, we can delve deep into the workings of LDAP and uncover its hidden treasures.

Most directories do not allow anonymous writing. To be able to add/modify/delete data, you need to bind to the directory server host with a valid user name and password. Most directories also do not allow anonymous read (or search). Depending on your environment and sensitivity, you may also wish to do away with anonymous read on your directory. This will depend on your *ldap-server* configuration.

Configuring the LDAP client

If you query a single directory server (within your organisation), typing the hostname and search base on the command line each time might seem like a mundane and irritating task. You can customise the */etc/openldap/ldap.conf* (on Fedora/RHEL) file as shown below to specify an LDAP server and base. The OpenLDAP client utilities read the */etc/openldap/ldap.conf* file.

Warning: Do not modify the */etc/ldap.conf* file. It is for a different purpose and is accessed by a different set of programs.

To be able to simply query the above server (*x500.bund.de*), just specify the following two lines in */etc/openldap/ldap.conf*:

```
HOST x500.bund.de
BASE o=bund,c=de
```

Make sure you comment out all the other lines at this stage. Once you have done this, just execute the *ldapsearch* command without the host and the base options:

```
[vbg@vbg-work scratch]$ ldapsearch -x
```